

CSCI 4050/6050

Software Engineering

Online Bookstore System

Deliverable 3: Domain Class Diagram

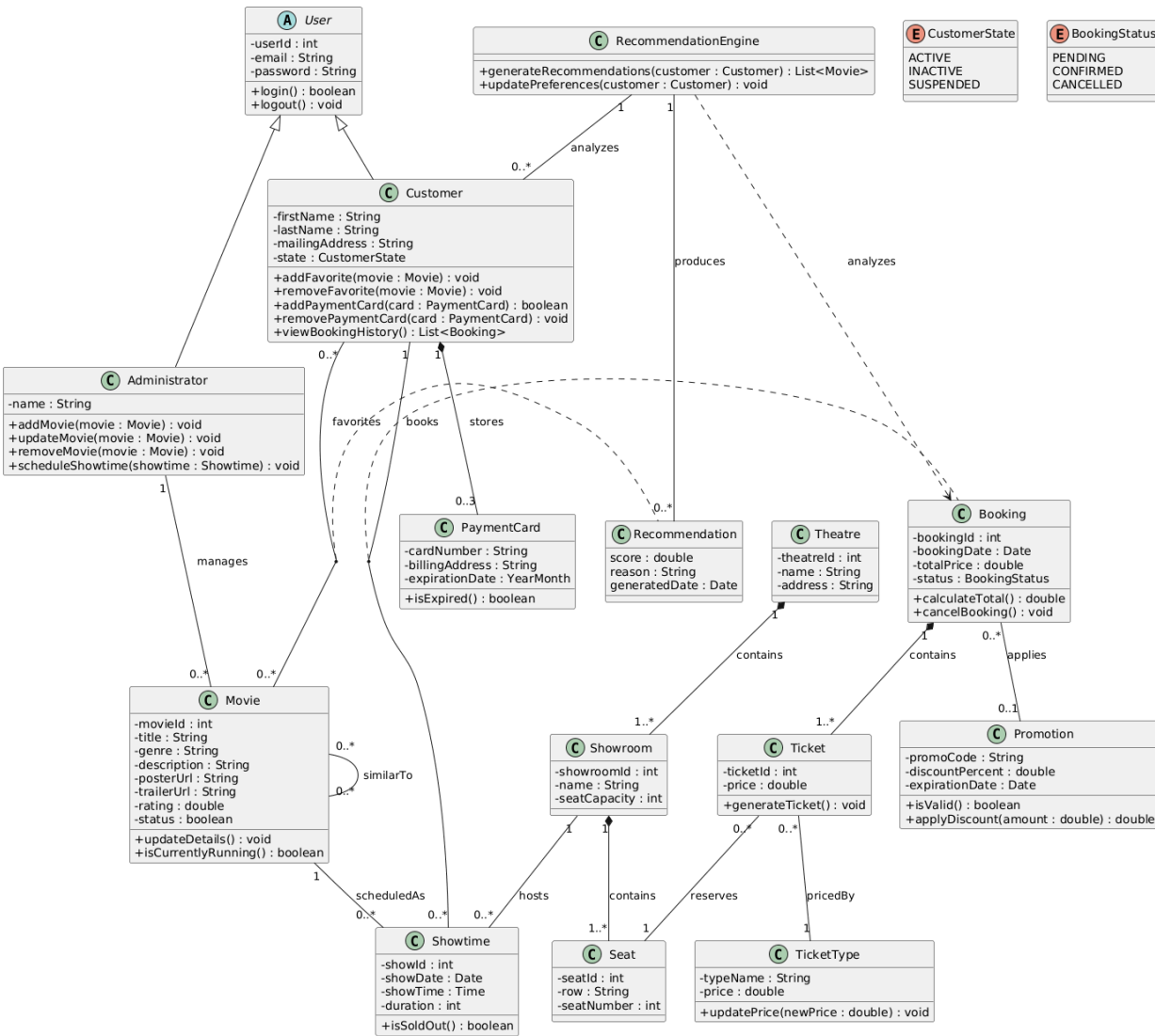
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Team 11

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Cinema E-Booking System Domain Class Diagram



Source code for PlantUML diagram:

```
@startuml
title Cinema E-Booking System Domain Class Diagram
skinparam classAttributeIconSize 0

' USERS -----
abstract class User {
    - userId : int
    - email : String
    - password : String
    + login() : boolean
    + logout() : void
}

class Customer {
    - firstName : String
    - lastName : String
    - mailingAddress : String
    - state : CustomerState
    + addFavorite(movie : Movie) : void
    + removeFavorite(movie : Movie) : void
    + addPaymentCard(card : PaymentCard) : boolean
    + removePaymentCard(card : PaymentCard) : void
    + viewBookingHistory() : List<Booking>
}

class Administrator {
    - name : String
    + addMovie(movie : Movie) : void
    + updateMovie(movie : Movie) : void
    + removeMovie(movie : Movie) : void
    + scheduleShowtime(showtime : Showtime) : void
}
```

```
enum CustomerState {
    ACTIVE
    INACTIVE
    SUSPENDED
}

' PAYMENT -----
class PaymentCard {
    - cardNumber : String
    - billingAddress : String
    - expirationDate : YearMonth
    + isExpired() : boolean
}

' MOVIE -----
class Movie {
    - movieId : int
    - title : String
    - genre : String
    - description : String
    - posterUrl : String
    - trailerUrl : String
    - rating : double
    - status : boolean
    + updateDetails() : void
    + isCurrentlyRunning() : boolean
}

' THEATRE STUFF -----
class Theatre {
    - theatreId : int
    - name : String
    - address : String
}
```

```
class Showroom {  
    - showroomId : int  
    - name : String  
    - seatCapacity : int  
}
```

```
class Seat {  
    - seatId : int  
    - row : String  
    - seatNumber : int  
}
```

```
' SCHEDULING -----  
class Showtime {  
    - showId : int  
    - showDate : Date  
    - showTime : Time  
    - duration : int  
    + isSoldOut() : boolean  
}
```

```
' BOOKING AND TICKETS -----  
class Booking {  
    - bookingId : int  
    - bookingDate : Date  
    - totalPrice : double  
    - status : BookingStatus  
    + calculateTotal() : double  
    + cancelBooking() : void  
}
```

```
enum BookingStatus {  
    PENDING  
    CONFIRMED  
    CANCELLED
```

```
}
```

```
class Ticket {  
    - ticketId : int  
    - price : double  
    + generateTicket() : void  
}
```

```
class TicketType {  
    - typeName : String  
    - price : double  
    + updatePrice(newPrice : double) : void  
}
```

```
class Promotion {  
    - promoCode : String  
    - discountPercent : double  
    - expirationDate : Date  
    + isValid() : boolean  
    + applyDiscount(amount : double) : double  
}
```

```
' RECOMMENDATION SYSTEM -----
```

```
class RecommendationEngine {  
    + generateRecommendations(customer : Customer) : List<Movie>  
    + updatePreferences(customer : Customer) : void  
}
```

```
class Recommendation {  
    - score : double  
    - reason : String  
    - generatedDate : Date  
}
```

```
' RELATIONSHIPS -----
```

User <|-- Customer

User <|-- Administrator

Customer "1" *-- "0..3" PaymentCard : stores

Customer "0..*" -- "0..*" Movie : favorites

Movie "0..*" -- "0..*" Movie : similarTo

Administrator "1" -- "0..*" Movie : manages

Theatre "1" *-- "1..*" Showroom : contains

Showroom "1" *-- "1..*" Seat : contains

Showroom "1" -- "0..*" Showtime : hosts

Movie "1" -- "0..*" Showtime : scheduledAs

Customer "1" -- "0..*" Showtime : books

(Customer, Showtime) .. Booking

Booking "1" *-- "1..*" Ticket : contains

Booking "0..*" -- "0..1" Promotion : applies

Ticket "0..*" -- "1" Seat : reserves

Ticket "0..*" -- "1" TicketType : pricedBy

RecommendationEngine "1" -- "0..*" Customer : analyzes

RecommendationEngine "1" -- "0..*" Recommendation : produces

(Customer, Movie) .. Recommendation

RecommendationEngine ..> Booking : analyzes

@endum1